SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks.* *(BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Short Answer Test

1. Suppose a user sends an email from a client system to a remote mail server through a network. The communication follows the OSI model, in which each layer performs specific functions to ensure the successful delivery of the message.
(a) Identify and explain the role of each OSI layer involved in this communication.
(b) Illustrate the process of data encapsulation across the layers for this scenario.
(c) Explain how the receiver reconstructs the original message using de-encapsulation.

2. Consider a scenario where a large file is transmitted from one computer to another over a network using reliable communication protocols.
(a) Explain how the Transport layer ensures reliable data transfer in this case.
(b) Describe the role of segmentation, flow control, and error control in this process.
(c) Illustrate how acknowledgments help in maintaining data integrity.

3. Suppose a data packet travels across multiple networks from source to destination using routers and switches.
(a) Explain the roles of the Network layer and Data Link layer in forwarding the packet.
(b) Describe how IP addressing and MAC addressing are used in this process.
(c) Illustrate how routing decisions are made at intermediate devices.

4. Consider a situation where errors occur during data transmission due to noise in the communication channel.
(a) Identify which OSI layers are responsible for error detection and correction.
(b) Explain how these layers work together to ensure reliable communication.
(c) Provide an example of how retransmission occurs when an error is detected.

5. Suppose a web browser sends a request to access a webpage using HTTP protocol.
(a) Explain how each OSI layer contributes to this web communication process.
(b) Illustrate the flow of data units (segment, packet, frame, bits) across layers.

(c) Describe how the response from the server reaches the client system.

6. Consider a network troubleshooting scenario where a system fails to communicate with another system.

(a) Explain how the OSI model can be used to identify the source of the problem.

(b) Apply a layer-by-layer approach to diagnose the issue.

(c) Suggest possible solutions based on the identified faulty layer.

7.Consider a scenario where data is transmitted between two devices in different networks using routers and switches.

(a) Explain how OSI layers interact with networking devices during transmission.

(b) Describe the flow of data across layers from sender to receiver.

(c) Analyze how addressing mechanisms ensure correct delivery across networks.

8.Consider a scenario where a real-time voice call (VoIP) is transmitted over a network.

(a) Explain how OSI layers support real-time data transmission.

(b) Describe how the data transmission process differs from file transfer.

(c) Analyze how layers coordinate to maintain continuous communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation

Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand